

ELECTIVE COURSE-9.2

UCS312 : Database Management System				
	L	T	P	Cr
	2	0	2	3.0
Course Objective: Emphasis is on the need of information systems. Main focus is on E-R diagrams, relational database, concepts of normalization and de-normalization and SQL commands.				
Syllabus Introduction: Data, data processing requirement, desirable characteristics of an ideal data processing system, traditional file-based system, its drawback, concept of data dependency, Definition of database, database management system, 3-schema architecture, database terminology, benefits of DBMS. Relational Database: Relational data model: Introduction to relational database theory: definition of relation, keys, relational model integrity rules. Database Analysis: Conceptual data modeling using E-R data model -entities, attributes, relationships, generalization, specialization, specifying constraints, Conversion of ER Models to Tables, Practical problems based on E-R data model. Database Design: Normalization- 1NF, 2NF, 3NF, BCNF, 4NF and 5NF. Concept of De-normalization and practical problems based on these forms. Database Implementation: Introduction to SQL, DDL aspect of SQL, DML aspect of SQL – update, insert, delete & various form of SELECT- simple, using special operators, aggregate functions, group by clause, sub query, joins, co-related sub query, union clause, exist operator. PL/SQL - cursor, stored function, stored procedure, triggers, error handling, and package.				
Laboratory Work Students will perform SQL commands to demonstrate the usage of DDL and DML, joining of tables, grouping of data and will implement PL/SQL constructs.				
Course Learning Outcomes (CLOs) The students will be able to: 1. Analyze the Information Systems as socio-technical systems, its need and advantages as compared to traditional file-based systems. 2. Analyze and design database using E-R data model by identifying entities, attributes and relationships. 3. Apply and create Relational Database Design process with Normalization and Denormalization of data. 4. Demonstrate use of SQL and PL/SQL to implementation database applications.				
Text Books				

Approved in 109th meeting of the Senate held on March 16, 2023. Revised in 112th and 113th meeting of the Senate held on March 11 and September 7, 2024, respectively. Further, revision Approved in 1st meeting of Academic Council held on September 11, 2025.

1. Database System Concepts, Silverschatz A., Korth F. H. and Sudarshan S., Tata McGraw Hill 6th ed, 2010.
2. Fundamentals of Database Systems, Elmasri R. and Navathe B. S., Pearson, 7th ed, 2016.

Reference Books

1. Modern Database Management, Pearson, Hoffer J., Venkataraman, R. and Topi, H., Pearson, 12th ed, 2016.
2. Simplified Approach to DBMS, Parteek Bhatia and Gurvinder Singh, Kalyani, 8th ed, 2004.

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